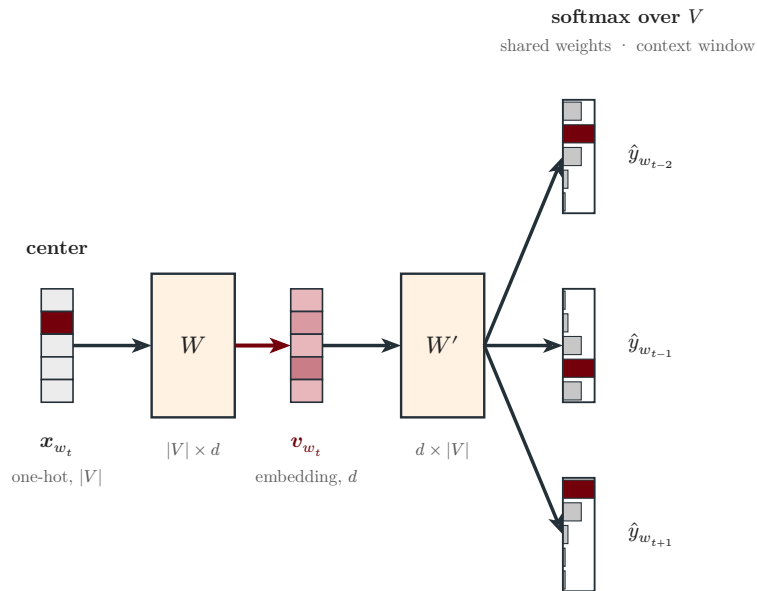


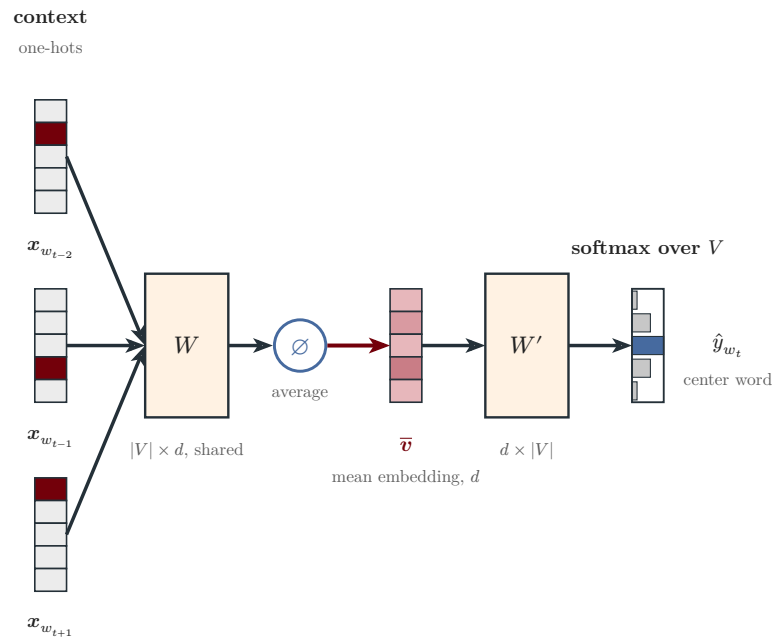
Skip-gram

predict each context word from the center word



CBOW

predict the center word from the averaged context



Both models are **shallow** — no hidden non-linearity. The projection matrix W ($|V| \times d$) is itself the learned word-embedding table: row w is the vector for word w . Skip-gram maximizes the probability of each surrounding word given the center; CBOW does the reverse on the averaged context. In practice the full softmax over $|V|$ is approximated by negative sampling or hierarchical softmax.